

# Sideline Screen / Duck Blind

## Construction and Field Operations Manual

The nominal 4 ft x 8 ft sideline screen provides the color guard with a concealed area for storing equipment and making costume or equipment changes. The frame is constructed from 3/4-in. EMT conduit, off-the-shelf brackets, and 3D-printed parts. Decorative vinyl covers the finished face. The completed screen folds flat for transport and storage and unfolds for field deployment.



Screen shown deployed and folded for storage.

Release v0 (unreleased) | Development build D27 | September 9, 2026

| Version | Date       | Status        | One-line description                                                           |
|---------|------------|---------------|--------------------------------------------------------------------------------|
| D24     | 2026-09-01 | Working draft | Imported legacy v24 draft as controlled baseline D24                           |
| D25     | 2026-09-01 | Working draft | Standardized status markings, footers, and added visible version history table |

| Version | Date       | Status        | One-line description                                                                                                   |
|---------|------------|---------------|------------------------------------------------------------------------------------------------------------------------|
| D26     | 2026-09-01 | Working draft | Standardized document architecture, 2026 CBA rules, transport cart specifications, and canonical cost breakdown tables |
| D27     | 2026-09-09 | Working draft | Add vinyl installation procedure, dimensions specifications, and tape/clip fastening                                   |

## Intro / Overview

This master manual is organized as two independently printable appendices under a shared overview. The appendix letter in each header and page number identifies the complete print unit.

### Appendix A: Operations Manual

Field-facing copy for transport, deployment, ballasting, post-use inspection, and storage. A field operator should need only the pages labeled A- once the procedures are validated and released.

### Appendix B: Construction Manual

Shop-facing copy for materials, cutting, frame assembly, vinyl installation, inspection, and source lookup. The component-source index is located at the end of Appendix B and applies only to that appendix.

### Printing and separation

**Appendix A print unit:** Print or extract every page labeled A-. Its cover identifies the parent manual and revision.

**Appendix B print unit:** Print or extract every page labeled B-. Its cover and index remain with the construction packet.

**Master copy:** Keep this Intro / Overview with the complete document so the relationship between both appendices remains clear.

**Working-draft status:** Appendix A is structurally ready to separate, but its operating procedures are still development notes and must not yet be issued for field use. Appendix B's frame-construction procedure is illustrated; its vinyl-installation procedure remains incomplete.

## APPENDIX A

# Operations Manual

Field transport, deployment, ballasting, inspection, and storage

This appendix is designed as a complete, independently printable work packet while remaining clearly identified as part of the parent manual.

**Audience:** Field operators

**Contents:** Folding and transport; deployment; ballasting; post-use inspection and storage

**Print or carry:** All pages labeled A-

Parent revision: Release v0 (unreleased) | Development build D27 | September 9, 2026

**WORKING DRAFT - NOT READY FOR FIELD ISSUE:** Deployment sequence, approved ballast quantity, and operating limits still require validation. Do not issue this appendix to field crews yet.

*When separated from the master document, keep this cover as the first page of Appendix A.*

## Field Operations

Appendix A is the field-facing portion of the manual. It is designed to be printed and carried independently by operators responsible for transport, deployment, inspection, ballasting, and storage.

**Section status:** The procedures below are development notes, not a complete field-use procedure. Deployment sequence, approved ballast quantity, and operating limits still require validation.

### Folding and transport

To prepare sideline screens for transport, handlers release the hinged arm clips, fold the bottom support arms (E) upward against the rear frame, and collapse the entire rear support frame flat against the main frame. Multiple folded screens are loaded vertically onto a dedicated rolling transport cart between side guide rails. Front and rear retaining gates are closed and latched (Cart Gate Latch) to lock screens in place during transit from the equipment trailer to the stadium entrance.

### Deployment and Sideline Cart Stowage

At the performance field gate, the transport cart is rolled to the sideline staging position. Handlers unlatch and open the front and back cart gates, lift individual folded screens from the cart bed, and carry them to their designated sideline coordinates. Once on coordinate, operators swing the rear support frame open and snap the hinged arm clips securely over the bottom support arms (E). During the performance, the empty transport cart remains parked and braked along the sideline outside the performance boundary. Immediately upon the final performance cue, screens are collapsed flat, rapidly reloaded into the transport cart, gates latched, and the cart rolled off the field in continuous movement per Rule 8.05.

### Ballasting

The current design allows ballast to be suspended from upper weight clips. Placing weights over the bottom support components may fatigue or break the 3D-printed hinged arm clips and is not an approved method until load testing is complete.

**Do not issue for field use:** Before this subsection is released, specify the approved weight clips, ballast quantity, attachment locations, wind limits, inspection criteria, and response to changing weather.

## Applicable 2026 Marching Band Competition & Prop Rules

The following governing rules are extracted directly from the official 2026 Colorado Bandmasters Association (CBA) Marching Band Rulebook (Appendix B Index [11]). All prop builders, parent crew, and student field operators must adhere to these standards:

**ANNUAL RULEBOOK NOTICE:** This section reflects the governing rules for the 2026 competitive season. Logistics staff and build leads must review this section annually against updated CBA and Bands of America (BOA) rulebooks to verify continuing compliance.

### Rule 8.05: Props, Equipment, Surface Protection, and Double-Bagging

- **Gate Ingress:** All props and equipment must be designed and of a quantity such that they can be brought onto the Performance Field from the band entrance gate.
- **Continuous Movement:** Following the end of the band's Performance, all props and equipment must be in continuous movement until entirely removed from the stadium.
- **Turf Contact Protection:** All wooden props must be protected with a heavy-duty sustainable plastic product (PVC, Melamine, etc.) where said prop comes in contact with the field surface. Steel props with smooth edges are acceptable without the need for additional protection.
- **Secondary Containment for Sandbags:** Props must not leave holes in the Performance Field surface. Sand bags must be in a secondary container (bucket, double bagged, etc.) that will prevent sand from leaking.

### Rule 8.07: Equipment Wheel and Turf Protection Standards

- **Pneumatic-Like Tires:** All instruments and equipment wheeled into the stadium from the entrance gate forward must have pneumatic-like tires. Wheels must be able to support the weight of the instrument/prop without creating damage to the field surface.

### Rule 8.08: Prop Staging Height Restrictions

- **12-Foot Rigid Height Limit:** Staging (props, backdrops, screens, or similar objects) built and/or used by bands at CBA sanctioned events shall be limited to a maximum total height of twelve (12) feet, including wheels, platforms, and safety railings. Materials such as wood, metal, plastic, or aluminum are not permitted above the twelve-foot limit. (Note: The Sideline Screen is a nominal 4 ft height when deployed, safely conforming to this rule).

### Rule 8.09: Prop Assembly, Timing, and USAFA Falcon Stadium Clearance

- **Staging Ingress:** A band may bring props into the stadium after the previous band has entered the field. Props must fit in the end zone curve without blocking entrance/exit gates.

- **Falcon Stadium 9'6" Tunnel Rule:** Any prop being used at Falcon Stadium (United States Air Force Academy) must not exceed 9 ft 6 in. to clear the tunnel entrance. Props up to the 12-ft limit may be erected in the tunnel after clearing the initial entrance.
- **Falcon Stadium Venue Update (2025/2026):** While the 9'6" tunnel restriction remains in the printed CBA rulebook, it has been rendered obsolete by stadium renovations completed prior to the 2025 season. The lower overhead crossbeam at the bottom of the Falcon Stadium tunnel was permanently removed, allowing props to clear without restriction.

#### **Rule 9.07: Parent Field Access & Wristband Limits**

- **25 Field Pass Wristbands:** Access to the field is restricted to Directors/Staff with CBA passes and parents assisting with props/front ensemble with designated Field Pass wristbands. Each band receives exactly 25 Field Pass wristbands upon check-in. Additional wristbands are not available.

#### **Post-use inspection and storage**

This subsection will cover ballast removal, damage checks, drying and cleaning, folding, transport, storage position, and reporting damaged brackets, conduit, vinyl, or printed parts.

## APPENDIX B

# Construction Manual

Materials, cutting, frame assembly, vinyl installation, and source lookup

This appendix is designed as a complete, independently printable work packet while remaining clearly identified as part of the parent manual.

**Audience:** Fabrication and installation crews

**Contents:** Components; tools; cutting; frame assembly; vinyl installation; Appendix B Index

**Print or carry:** All pages labeled B-

Parent revision: Release v0 (unreleased) | Development build D27 | September 9, 2026

**WORKING DRAFT - VINYL PROCEDURE PENDING:** The illustrated frame-construction procedure is included. The vinyl-installation procedure remains reserved for a future development pass.

*When separated from the master document, keep this cover as the first page of Appendix B.*

## Construction

Appendix B is for the fabrication and installation crew. It covers one complete frame and reserves a separate section for vinyl installation. Field operations are maintained independently in Appendix A.

**Orientation rule:** The finished face is the side that will be covered by decorative vinyl. Keep self-drilling screw heads, T-bracket nuts, bolt tails, and other sharp projections on the rear face unless a step specifically directs otherwise.

## Safety and work area

1. Wear eye protection when cutting conduit or driving self-drilling screws. Wear gloves while handling cut conduit.
2. Cut ends can be sharp. Deburr every cut before assembly.
3. Assemble the frame on a large, flat surface, such as a garage floor.
4. Check the frame for square frequently. The brackets allow some movement, so correct misalignment when practical without forcing the assembly.
5. After the outer frame is complete, install each successive rail or bracket toward the center of the frame, immediately inside the previously installed component.

## Materials and components

Purchase seven 10-ft lengths of 3/4-in. EMT conduit for one screen. Cut them into components A-E using the optimized cut plan below. The Index ref. column links each component to its purchase source, print material, or digital part files in the Appendix B Index at the end of this appendix. The illustrated inventory that follows provides a visual identification check for components A-J.

| ID | Qty. | Component             | Specification            | Index ref.        |
|----|------|-----------------------|--------------------------|-------------------|
| A  | 4    | Long main-frame rails | 92.5-in. EMT conduit     | <a href="#">1</a> |
| B  | 2    | Main-frame end rails  | 44.5-in. EMT conduit     | <a href="#">1</a> |
| C  | 1    | Rear cross rail       | 86.5-in. EMT conduit     | <a href="#">1</a> |
| D  | 2    | Rear-frame arms       | 37-in. EMT conduit       | <a href="#">1</a> |
| E  | 2    | Bottom support arms   | 27.5-in. EMT conduit     | <a href="#">1</a> |
| F  | 6    | Corner brackets       | Purchased clamp brackets | <a href="#">2</a> |

| ID | Qty.   | Component                        | Specification                                     | Index ref.                            |
|----|--------|----------------------------------|---------------------------------------------------|---------------------------------------|
| G  | 8      | T-brackets                       | Purchased clamp brackets; four serve as hinges    | <a href="#">3</a>                     |
| H  | 6      | Corner plugs                     | 3D printed in black ASA                           | <a href="#">5</a> , <a href="#">6</a> |
| I  | 2      | Hinged arm clips                 | 3D printed in black ASA                           | <a href="#">5</a> , <a href="#">7</a> |
| J  | 36     | Self-drilling sheet-metal screws | #8 x 1/2-in. external-hex head                    | <a href="#">4</a>                     |
| -  | 1      | Custom vinyl banner              | Heavyweight outdoor scrim vinyl (dimensions TBD)  | Custom                                |
| -  | 1 roll | Double-sided tape                | 1-in. heavy-duty (16x ~4-in. strips per screen)   | <a href="#">9</a>                     |
| -  | 16     | Greenhouse snap clamps           | 1-in. pipe clamp clips (for 3/4-in. EMT w/ vinyl) | <a href="#">8</a>                     |

Illustrated component inventory

Match the letter on each drawing to the ID in the component table before cutting or assembly.

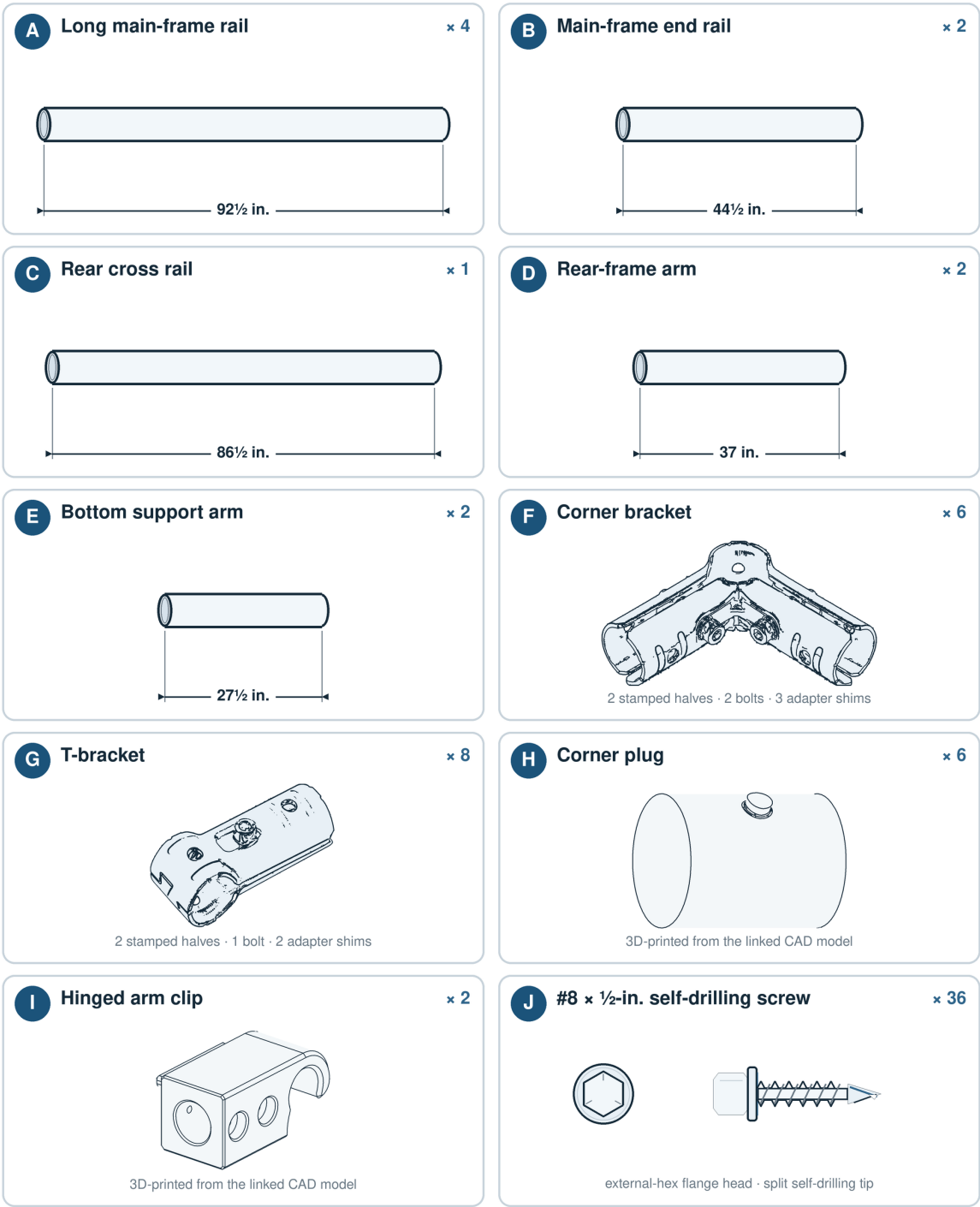


Figure 1. Component inventory for one complete frame.

Identification note: Drawings are not measuring templates and items are not shown at a common scale. The written dimensions and quantities control.

## Tools

6. EMT conduit cutter
7. Tape measure and permanent marker
8. Framing square or a tape measure long enough to compare frame diagonals
9. Deburring tool or metal file
10. 5 mm Allen key
11. Drill/driver with a 3/8-in. nut-driver bit
12. Eye protection and work gloves
13. 3D printer capable of printing ASA, unless the printed parts are prepared in advance

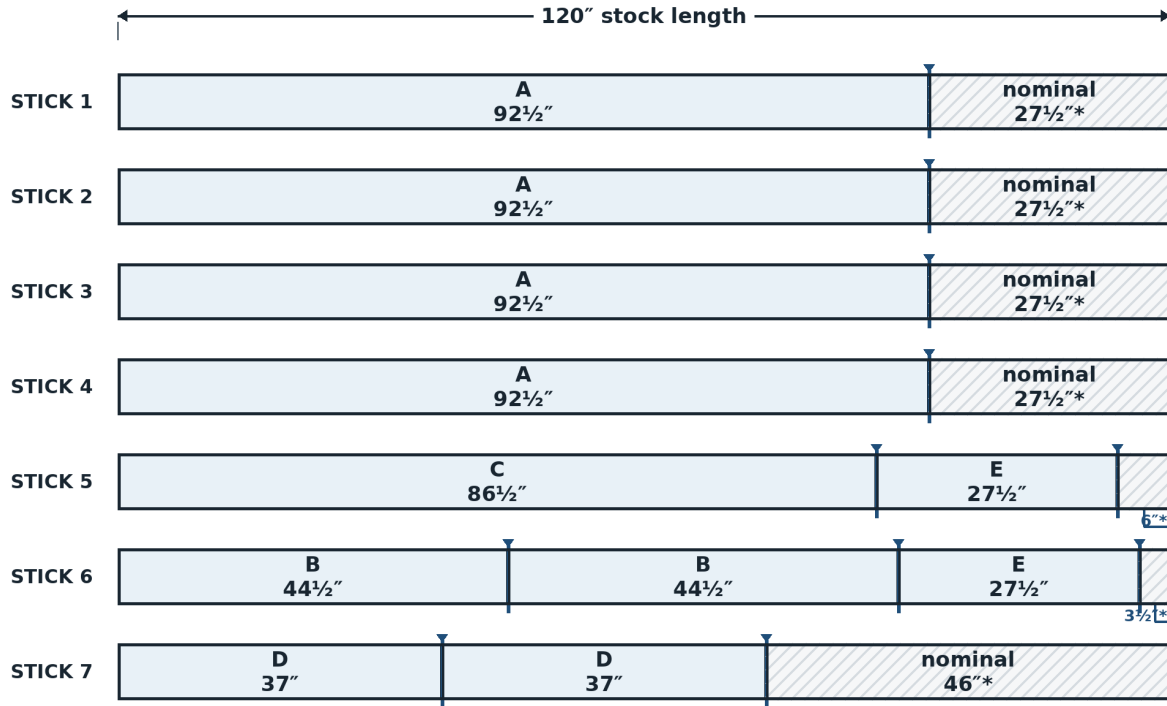
## 3D-printed parts

Black ASA is the recommended material. Polymaker ASA has provided adequate strength and UV stability in band-prop service for three seasons. PETG may be substituted, but strength varies substantially by brand and formulation; print and load-test a sample before producing all structural parts. Previous Sunlu PETG feather bases fractured in service.

**Material restriction:** Do not use PLA or ABS for these parts because neither is suitable for prolonged outdoor UV exposure.

## Optimized conduit cut plan

The following plan produces every required tube from seven 10-ft stock lengths and does not depend on the nominal 27.5-in. remainder from a 92.5-in. cut. This leaves allowance for normal cutting loss. Keep every finished piece within 1/8 in. of its specified length.



\* NOMINAL OFFCUT BEFORE CUTTING LOSS — SEE CUTTING NOTE BELOW

Figure 2. Proportional seven-stick conduit cut diagram.

**Cutting note:** Gray areas are nominal remainders before cutting loss. Measure every retained piece from a clean stock end. Do not use the nominal 27.5-in. remainder after an A cut as an E arm.

### Detailed cut list

| Stock length | Retained cuts                       | Approximate remainder        |
|--------------|-------------------------------------|------------------------------|
| 1            | 1 x 92.5 in. (A)                    | About 27.5 in.; spare/offcut |
| 2            | 1 x 92.5 in. (A)                    | About 27.5 in.; spare/offcut |
| 3            | 1 x 92.5 in. (A)                    | About 27.5 in.; spare/offcut |
| 4            | 1 x 92.5 in. (A)                    | About 27.5 in.; spare/offcut |
| 5            | 1 x 86.5 in. (C) + 1 x 27.5 in. (E) | 6 in., less cutting loss     |
| 6            | 2 x 44.5 in. (B) + 1 x 27.5 in. (E) | 3.5 in., less cutting loss   |
| 7            | 2 x 37 in. (D)                      | 46 in., less cutting loss    |

### Cutting procedure

1. Label the seven 10-ft stock lengths 1-7.
2. Mark every retained cut before cutting. Verify the complete cut plan before starting.
3. Cut each tube to length. Measure the finished piece rather than assuming an offcut is exact.
4. Deburr the inside and outside edges of every cut.
5. Label the finished tubes A-E so they can be identified during assembly.

When building multiple screens, pool and label usable offcuts before purchasing additional stock.

### Estimated fabrication cost breakdown

The following tables itemize estimated material costs per sideline screen and for the planned rolling transport cart based on standard off-the-shelf retail hardware pricing (excluding custom printed vinyl):

#### Per Sideline Screen / Duck Blind (Excl. Vinyl)

| Component / Category | Items Included                                                 | Est. Cost |
|----------------------|----------------------------------------------------------------|-----------|
| Conduit Framing      | 7x 3/4" x 10-ft EMT conduit sticks (Home Depot SKU 0550110000) | \$40.50   |
| Corner Brackets (F)  | 6x 3/4" 3-way corner brackets (Amazon B0D5BBKCKG)              | \$27.00   |
| T-Brackets (G)       | 8x 3/4" EMT T-brackets (Amazon B0CKQW11SJ)                     | \$23.00   |
| Fasteners (J)        | 36x #8 x 1/2" hex flange self-drilling screws                  | \$4.00    |
| 3D-Printed Parts     | 6x Plugs H, 2x Clips I, 2x Weight Clips in Black ASA           | \$10.00   |

| Component / Category    | Items Included                | Est. Cost        |
|-------------------------|-------------------------------|------------------|
| <b>TOTAL PER SCREEN</b> | Complete folding screen frame | <b>~\$105.00</b> |

### Sideline Screen Transport Cart (Component 2 Estimate)

| Subsystem / Category            | Items Included                                                            | Est. Cost        |
|---------------------------------|---------------------------------------------------------------------------|------------------|
| Rolling Base Chassis            | Backdrop-spec 2x4 framing, 1/2" plywood deck, 4x casters w/ brakes, paint | \$155.00         |
| Side Guide Rails                | Vertical framing / conduit dividers to hold screens upright               | \$35.00          |
| Front & Rear Retention Gates    | Hinged gate framing, strap/barrel hinges, and 3D-printed latches          | \$30.00          |
| <b>TOTAL PER TRANSPORT CART</b> | Complete rolling transport cart                                           | <b>~\$220.00</b> |

Frame assembly

Complete the four stages in order. At every hinge location, fasten the bracket to the short moving tube only; never drive a screw into the long rail around which the bracket must rotate.

Folded-state connection map

Use Figure 3 to identify the finished folded arrangement, then use Figure 3A to see how the individual members and connectors align before beginning Stage 1. Both drawings use the same component labels and connector color key. Tube width and separation distances are enlarged for clarity.

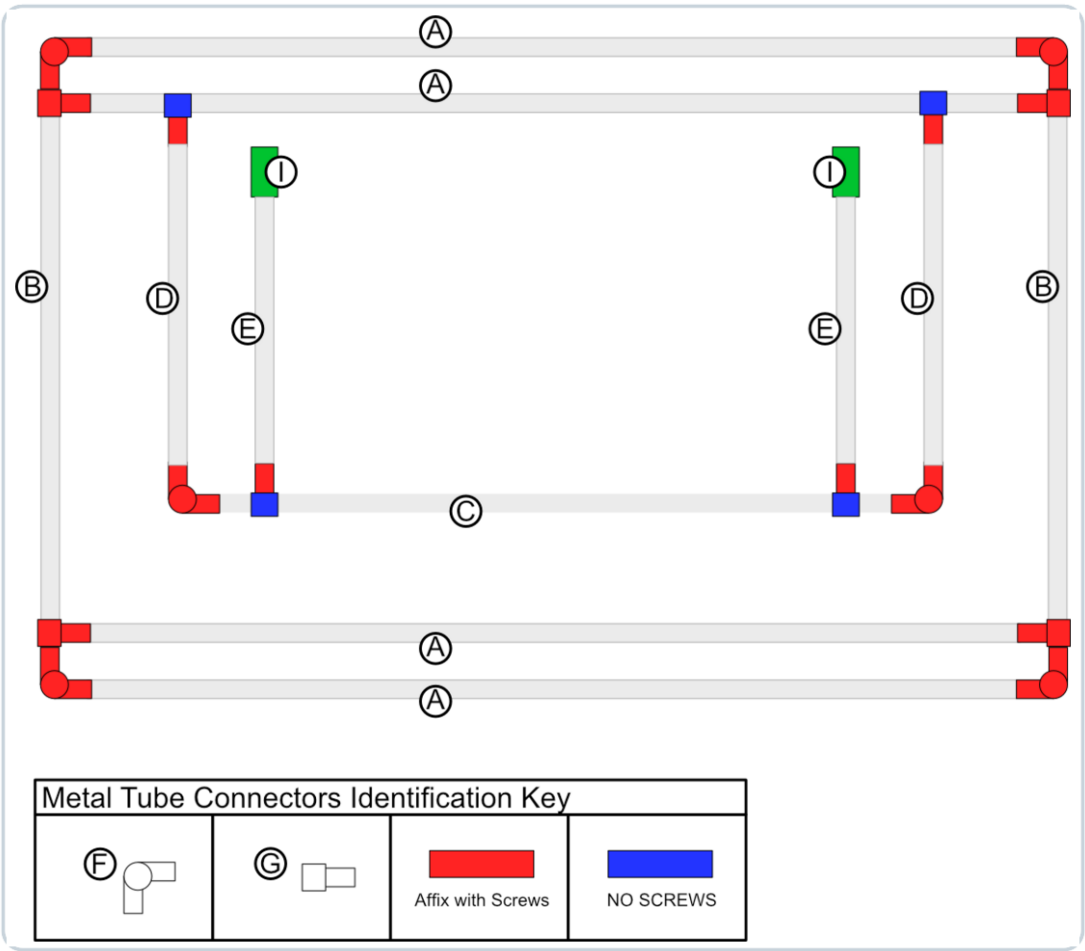


Figure 3. Folded-state component and connector-fastening map for the finished frame.

## Exploded folded-state component view

Use this exploded view to identify every construction component in the folded assembly. The separated member ends show how the A-E tubes align with the F and G connectors. Red markers identify connections that receive Item J screws, blue markers identify interfaces that must remain unfastened to pivot, and green blocks identify the two Item I hinged arm clips. Individual Item J fasteners are intentionally omitted.

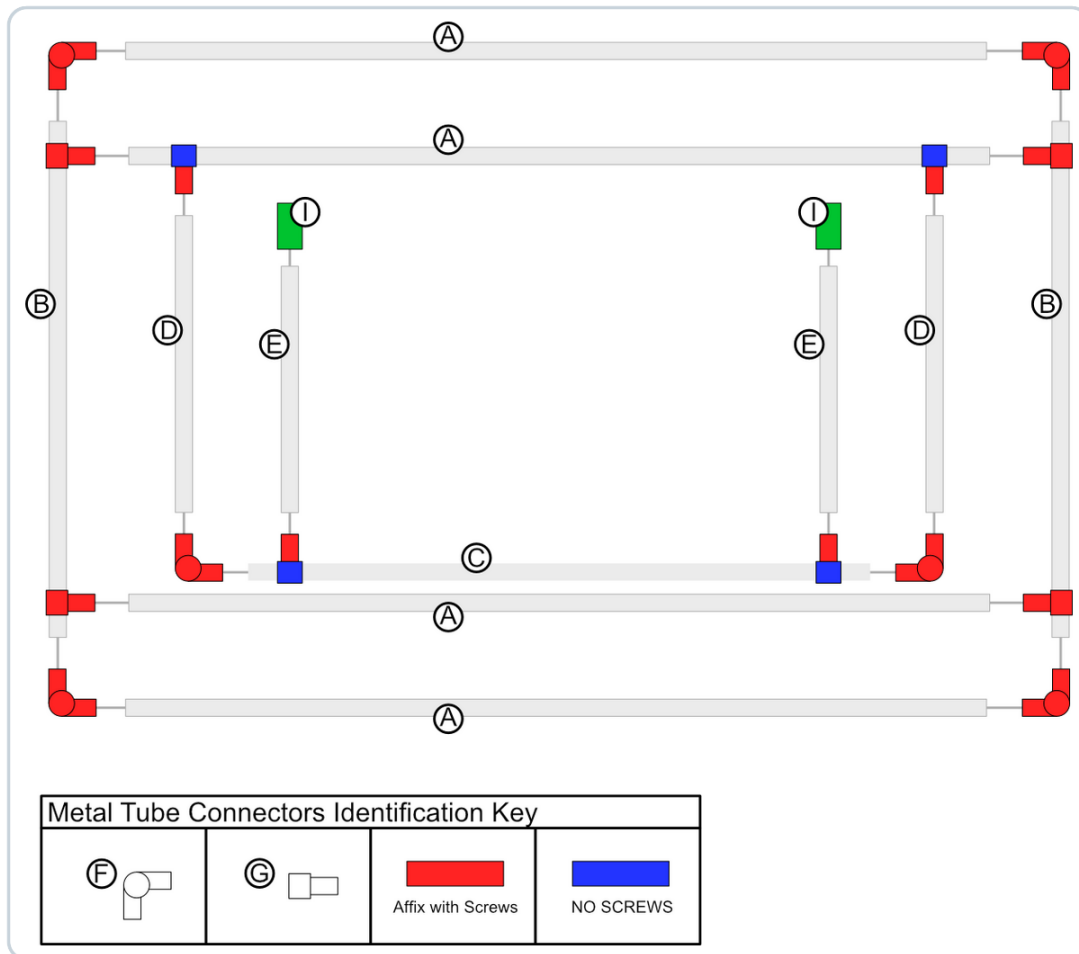


Figure 3A. Exploded folded-state component and connector-fastening map.

## Connection rules

**Affix with screws** — At every red connector marker, secure the connection from the inside or rear face with Item J self-drilling screws, using the quantity and locations specified in the applicable build stage.

**No screws** — At every blue connector marker, do not drive an Item J screw into the tube that must remain free to pivot. Install the specified screws only on the red-marked fixed side of the connector.

**Protect the vinyl** — Orient every Item J screw head, T-bracket nut, and bolt tail toward the inside or rear of the assembly, away from the finished face and vinyl.

## Stage 1: Assemble the main frame

### Parts for this stage

6. 2 x A rails
  7. 2 x B rails
  8. 4 x F corner brackets
  9. 4 x H corner plugs
  10. 8 x J screws
1. Lay two A rails and two B rails on the flat work surface to form a rectangle.
  2. Join the tubes with four F corner brackets. Insert one H corner plug into each unused third socket, rotating it so the side peg faces the rear and aligns with the hole on the bracket's outside bend.
  3. Measure both diagonals. Adjust the frame until the diagonal measurements are equal, then tighten all bracket-retaining bolts.
  4. Install one J screw at each of the eight tube-to-bracket joints. Drive the screws from the inside edge of the rectangle so no screw head faces the finished face.

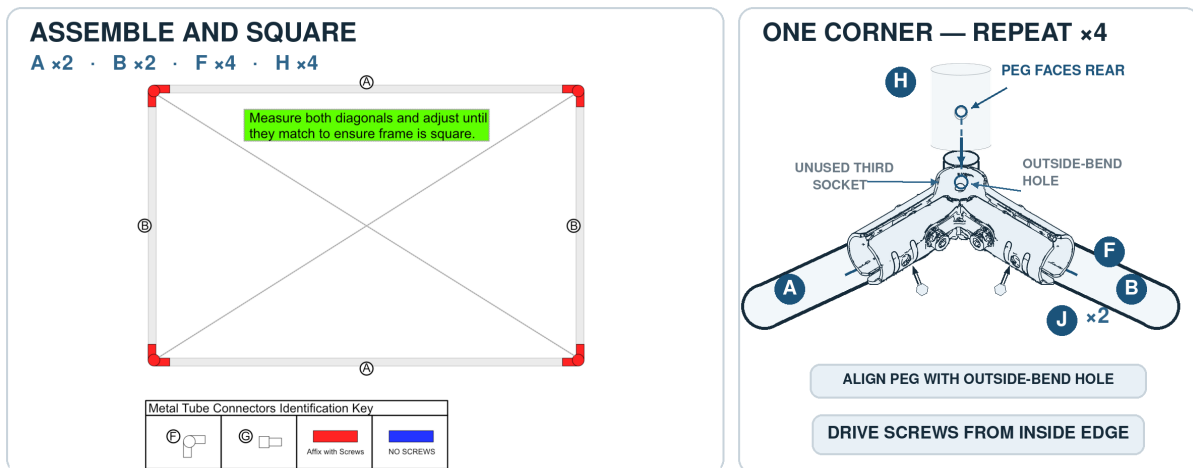


Figure 4. Stage 1 main-frame assembly and corner detail.

## Stage 2: Install the inner cross rails

### Parts for this stage

5. 2 x A rails
6. 4 x G T-brackets
7. 8 x J screws

**Fastener orientation:** Place the T-bracket nuts and bolt tails on the rear face. The front/finished face must remain free of sharp projections that could damage the vinyl.

1. Place one A rail immediately inside the top A rail and the other A rail immediately inside the bottom A rail.
2. Attach each end of both A rails with a G T-bracket. Insert each bracket bolt from the finished face toward the rear so the nut is on the rear face, then tighten fully.
3. Flip the main frame so the rear face is up.
4. Drive one J screw through each of the eight tube-to-T-bracket interfaces from the rear face.

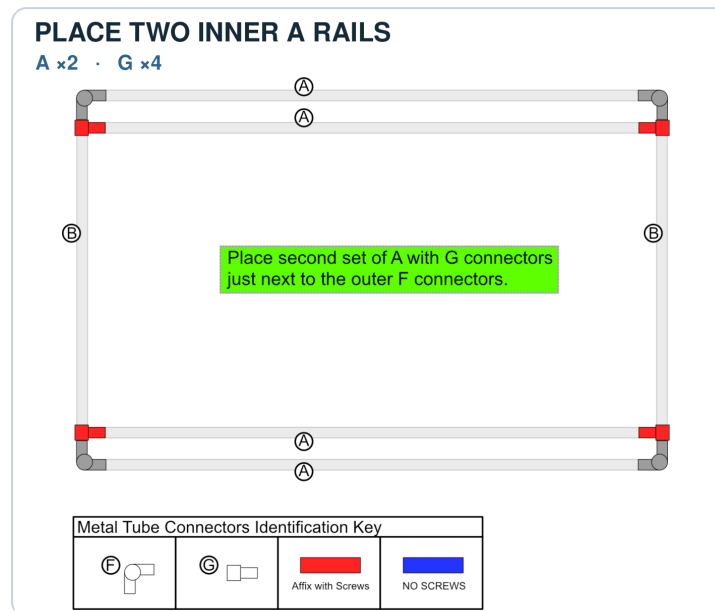


Figure 5. Stage 2 inner-rail placement and T-bracket fastening.

### Stage 3: Assemble and attach the rear support frame

#### Parts for this stage

5. 2 x D arms
6. 1 x C rear cross rail
7. 2 x F corner brackets
8. 2 x G T-brackets
9. 2 x H corner plugs
10. 8 x J screws

**Critical hinge rule:** Each G bracket in this stage must pivot around the upper A rail. Install two screws in each D arm, but no screws in the A rail.

1. Flip the main frame so the finished face is up. No additional self-drilling screws will be driven through the finished face.
2. Choose either long edge as the top and mark it. The upper inner cross rail is the second A rail from the top.
3. Attach one D arm near each end of the upper inner cross rail with a G T-bracket. Position each new bracket immediately inside the previously installed bracket at that end.
4. At each G T-bracket, drive two J screws into the D arm only (four screws total). Do not drive screws into the A rail; the bracket must rotate freely around that rail.
5. Join the free ends of the two D arms with the C rear cross rail, two F corner brackets, and two H corner plugs.
6. Square the rear support frame and tighten both corner brackets.
7. Install two J screws at each rear-frame corner (four screws total), placing them on the inside of the corner. Stage total: eight screws.

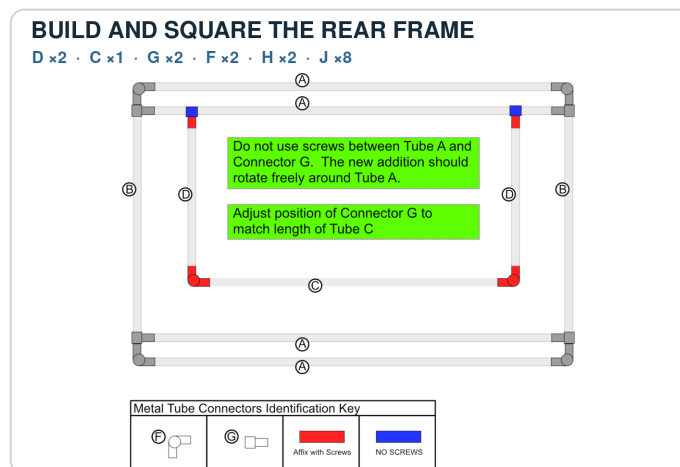


Figure 6. Stage 3 rear-support-frame assembly and hinge fastening.

## Stage 4: Install the bottom support arms

### Parts for this stage

8. 2 x E support arms
9. 2 x G T-brackets
10. 2 x I hinged arm clips
11. 12 x J screws

**Critical hinge rule:** Each G bracket in this stage must pivot around the C rear cross rail. Install screws in the E arms only; do not fasten either bracket to the C rail.

1. Attach one E support arm near each end of the C rear cross rail with a G T-bracket. Position each new bracket immediately inside the adjacent corner bracket and tighten the bracket bolts.
2. At each G T-bracket, drive two J screws into the E arm only (four screws total). Do not drive screws into the C rail; the bracket must rotate freely around that rail.
3. Stand the assembly upright with both E support arms lying on the ground.
4. Fit one I hinged arm clip onto the free end of each E arm. Engage both clips with the lower inner cross rail, which is the third long rail from the top and the second from the bottom.
5. Make each clip square to the lower inner cross rail and confirm that it is fully seated on its E arm.
6. Drive four J screws through each hinged arm clip into its E arm (eight screws total). Start with the accessible outside pair, disengage the clip, and install the remaining two screws. Stage total: twelve screws.

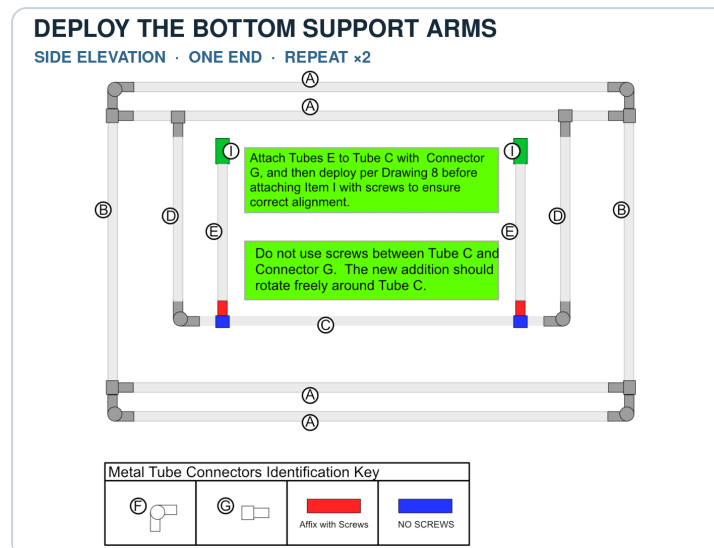


Figure 7. Stage 4 bottom-support-arm installation and clip fastening.

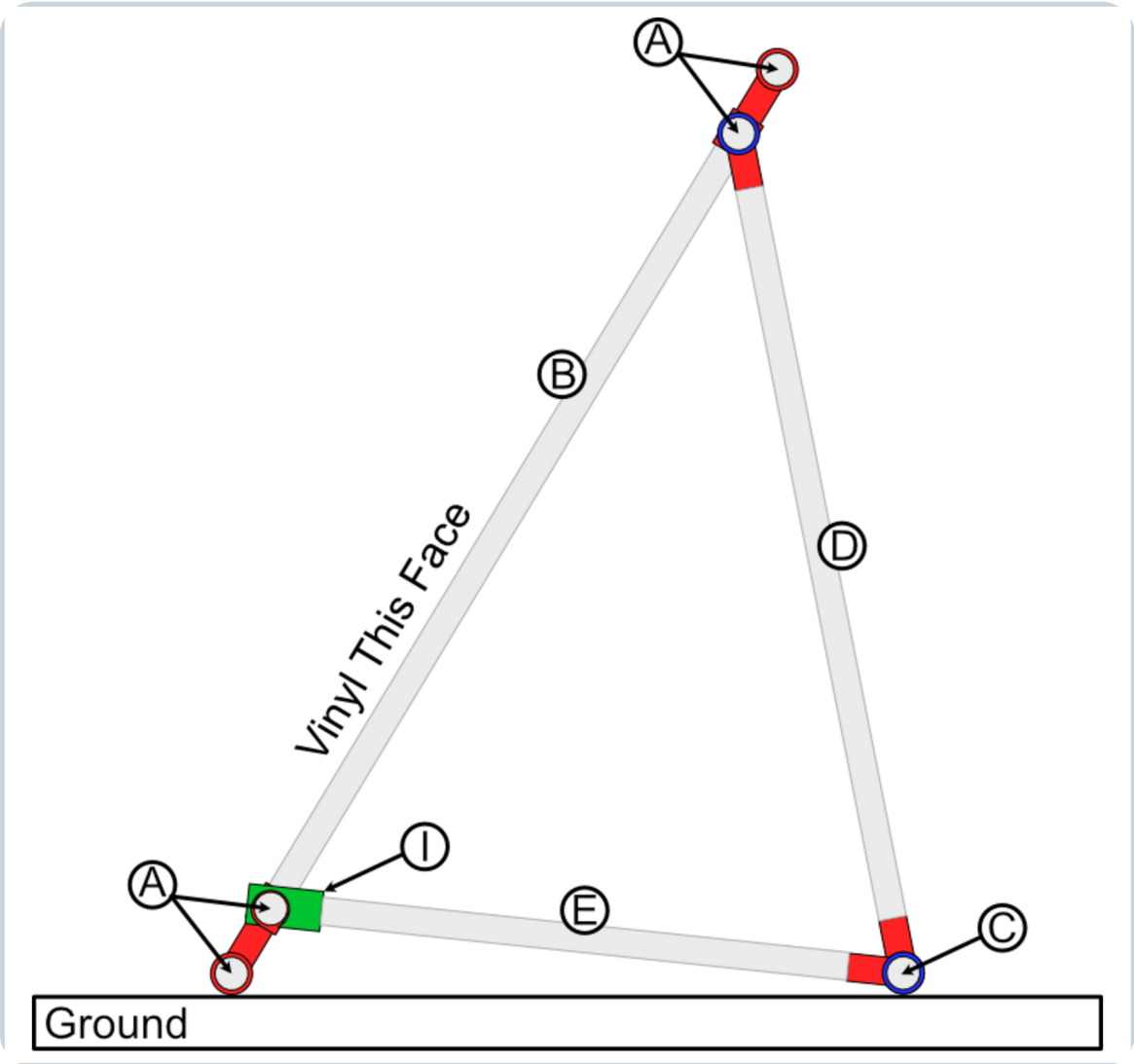


Figure 8. Deployed cutaway side elevation showing the B foot and C/E connection.

## Final construction inspection

7. All 36 self-drilling screws are installed in the specified locations.
8. No screw head, bracket nut, bolt tail, burr, or other sharp projection faces the future vinyl surface.
9. Both rear-frame hinges rotate around the upper inner rail without binding.
10. Both bottom-support hinges rotate around the rear cross rail without binding.
11. Both hinged arm clips engage and disengage without excessive force.
12. All retaining bolts are tight, and every structural corner is as square as practical.
13. The frame can be unfolded, locked open with both clips, released, and folded flat.

## Transport Cart Fabrication (Planned Component)

To support efficient field transport and rapid deployment during competitive shows, a dedicated rolling transport cart will be constructed to hold multiple folded sideline screens / duck blinds:

- **Backdrop Cart Base Compatibility:** The transport cart utilizes the same heavy-duty 96.0 in. x 44.5 in. 2x4 lumber base framing, 1/2-in. plywood decking, and four heavy-duty swivel plate casters standardized for the Rolling Backdrop System. Building on this shared chassis ensures dimensional consistency and uniform trailer nose wall loading.
- **Side Guide Rails:** Vertical guide rails are mounted along both the left and right longitudinal edges of the cart bed to contain folded sideline screens standing upright in parallel slots during rolling transit.
- **Front & Rear Latching Gates:** Hinged retention gates are installed across the front and rear openings of the cart. The gates are secured with 3D-printed Cart Gate Latches (Cart Gate Latch.FCStd) to lock screens firmly inside the cart bed while allowing quick release and unobstructed unloading at the field sideline.
- **Status:** Provisional design specification. Detailed CAD models, cut lists, and fastener schedules for the transport cart modifications will be finalized in subsequent build releases following prototype testing.

## Vinyl Installation

**Section status:** Working draft procedure for vinyl installation using perimeter double-sided tape and mechanical greenhouse snap clamps. Recommended vinyl dimensions (image size, bleed size, overall cut size) and tape strip quantities are provisional and subject to shop fit-up confirmation.

This section governs the preparation, alignment, dual-axis tensioning, and mechanical fastening of the custom printed vinyl coverings to completed sideline screen frames. Installing vinyl on the folding frame requires a clean staging area, careful bleed centering, double-sided tape adhesion, and mechanical clamping with greenhouse snap clamps to prevent loosening under high field wind loads.

### Recommended Vinyl Dimensions & Ordering Specifications

Sideline screen face coverings are custom-ordered commercial vinyl banners. Because the screens are positioned directly along the field boundary facing stadium audiences and adjudicators, the printed artwork must be oriented right-reading and centered precisely across the outer EMT conduit frame.

| Dimension Parameter                    | Recommended Value                   | Engineering & Fit Specification                                                                   |
|----------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Nominal Frame Outer Bounds</b>      | 48.0 in. H x 96.0 in. W             | Nominal outer dimensions of perimeter 3/4-in. EMT rectangular frame (tubes A and B).              |
| <b>Live Image Size (Visual Area)</b>   | <b>TBD</b> (Nominal ~46" x ~94")    | Front face graphic area visible to audience and judges; centered within outer tubing perimeter.   |
| <b>Bleed Allowance (Wrap Margin)</b>   | <b>TBD</b> (Nominal ~2.5" per edge) | Extended background bleed color wrapped around 3/4-in. EMT conduit to adhere to rear tape strips. |
| <b>Overall Flat Cut Size</b>           | <b>TBD</b> (Nominal ~53" x ~101")   | Total flat dimensions of ordered banner before edge wrapping, tensioning, and clamp installation. |
| <b>Substrate Material &amp; Weight</b> | Heavyweight scrim vinyl             | 13 oz. or 18 oz. tear-resistant outdoor banner vinyl with non-glare matte finish.                 |

| Dimension Parameter          | Recommended Value  | Engineering & Fit Specification                                                                                        |
|------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------|
| Edge Finishing Specification | Flush straight cut | Square raw-cut perimeter; DO NOT order hemmed borders or brass grommets, as raw edges wrap smoothly under snap clamps. |

**Dimensional specification note:** Exact custom dimensions—including visual image size, wrap-around bleed margin, and overall cut size—are currently provisional (TBD). Specific numerical values will be updated in the technical specifications following prototype field stretch testing.

## Required Materials & Fastening Hardware

Assemble all necessary staging items, adhesive consumables, and retention hardware before initiating vinyl mounting on a frame:

- **Moving blankets:** Two to three clean, heavy-duty moving blankets or carpet runners spread across the shop floor to cushion the printed vinyl face against dust, grit, and concrete abrasion.
- **Custom vinyl banner:** One custom printed outdoor banner per screen, rolled with print side facing inward. Unroll and inspect for print flaws and correct orientation prior to assembly.
- **Double-sided tape:** 1-in. wide heavy-duty outdoor mounting tape (Appendix B Index [9]). Cut into sixteen (16) strips of approximately 4-in. length (~64 linear inches per screen). Application schedule: 5 strips on each long rail (10 total) and 3 strips on each short rail (6 total). Tape quantities are provisional (TBD) and will be validated during build testing.
- **Greenhouse snap clamps:** Sixteen (16) 1-in. greenhouse clamp clips (Appendix B Index [8]). Snapped directly over the wrapped vinyl at every tape location (5 along each long rail, 3 along each short rail). Note: While the frame uses 3/4-in. trade size EMT conduit, EMT has an actual outer diameter of approximately 0.922 in. (nearly 1 in. OD); wrapped with vinyl and tape, 1-in. greenhouse snap clamps provide the proper snug snap fit.
- **Surface preparation (Optional):** Isopropyl alcohol (IPA) and clean microfiber cloth. Degreasing the conduit is completely optional—the heavy-duty tape combined with mechanical greenhouse snap clamps provides more than enough holding power. A weaker initial adhesive bond to the bare conduit is actually preferable, as it significantly simplifies tape removal and cleanup later.
- **Personnel:** Two (2) persons minimum for staging and long-side wrapping; three (3) persons recommended during short-side tensioning so opposite ends can be pulled simultaneously.

## Step-by-Step Vinyl Installation Procedure

### Step 1: Staging and Surface Protection

Thoroughly sweep and vacuum a clean, flat floor area measuring at least 10 ft x 14 ft. Lay moving blankets flat across the entire work zone to prevent grit from scratching the printed graphic.

Carefully unroll the custom vinyl banner face down directly on top of the moving blankets. Inspect the reverse side for cleanliness and smooth out any major packing rolls.

### Step 2: Frame Placement and Bleed Alignment

Carry the assembled sideline screen frame (in folded transport configuration) onto the vinyl and place it face down centered over the material.

Shift the frame until there is an even, equal reveal of perimeter bleed wrap margin on all four sides (top, bottom, left, and right outer rails). Verify that the frame is resting completely square on the vinyl prior to applying any tape.

### Step 3: Perimeter Double-Sided Tape Application

Cut sixteen (16) strips of 1-in. heavy-duty double-sided tape, each approximately 4 inches in length.

Apply the tape strips to the rear-facing surface of the perimeter EMT conduit in the following layout:

- **Long rails (Top and Bottom A rails):** Apply 5 strips spaced evenly along each rail (one centered, two at intermediate quarter points, and two approximately 3 inches inward from the corner brackets; 10 strips total).
- **Short end rails (Left and Right B rails):** Apply 3 strips evenly spaced along each rail (one centered and two approximately 3 inches inward from the corner brackets; 6 strips total).

Burnish each tape strip firmly onto the conduit surface, then peel and discard the protective release liners from all 16 strips.

### Step 4: Long-Side Tensioning and Initial Wrap

Start along one designated long side (typically the top A rail). Beginning at the center tape strip and progressing outward toward both corners, fold the vinyl bleed margin tightly around the conduit and press it firmly into the 5 double-sided tape strips on the rear of the tube. Check that the graphic remains aligned with the outer tubing edge.

Move to the opposing long side (bottom A rail). Starting at the center and working outward, pull the vinyl firmly across the frame face to eliminate all slack and wrinkles. Fold the vinyl edge tightly around the bottom conduit and press it securely into the 5 opposing tape strips.

Sight down the front face beneath the frame: confirm that longitudinal tension is smooth and uniform, with no wave patterns or diagonal wrinkling.

### Step 5: Short-Side Dual-Operator Tensioning and Wrap

Station one person at each of the two short end rails (B rails).

Working simultaneously, both operators firmly grasp the vinyl bleed margin on opposite ends and pull outward in opposing directions away from the center. This synchronized outward pull establishes balanced transverse tension across the entire width of the screen without pulling the frame out of square.

While maintaining continuous outward tension, wrap the vinyl edges over the short end rails and press them firmly into the 3 tape strips on each side.

Neatly fold the excess vinyl corner tabs around the corner brackets (F), flattening the material to maintain clean, square corners without bulky bunches.

### Step 6: Greenhouse Snap Clamp Installation

Install the 1-in. greenhouse snap clamps directly over the wrapped vinyl at every tape location, pressing each clamp firmly over the conduit until it snaps into full mechanical lock:

- **Long rails:** Install 5 snap clamps on the top rail and 5 snap clamps on the bottom rail (10 clamps total).
- **Short end rails:** Install 3 snap clamps on each of the two short end rails (6 clamps total).

The 1-in. greenhouse snap clamps compress the vinyl tightly against the tape and 3/4-in. EMT conduit, providing heavy-duty mechanical clamping that prevents tape peel or edge detachment under stadium wind gusts or rough trailer transit.

Confirm that no snap clamp interferes with the pivoting rear-support G-brackets, bottom support arm clips (I), or corner plugs (H).

### Step 7: Final Tension and Mechanical Clearance Inspection

With two operators, carefully lift the completed sideline screen upright off the moving blankets.

Inspect the front vinyl face in bright lighting: verify that the graphic is drum-tight, square, free of wrinkles or sagging, and centered within the visible frame opening.

Verify that all 16 greenhouse snap clamps are fully seated and firmly gripping the conduit over each tape strip.

Deploy the rear support frame (C/D) and engage both bottom support arm clips (I) into the lower inner rail (Figure 8). Confirm that the wrapped vinyl and snap clamps do not bind, rub, or impede frame deployment, clip latching, or folding flat for transport.

## Vinyl Removal & End-of-Season Teardown Protocol

**CAUTION — Temperature Requirement for Vinyl Removal:** Do not attempt removal of the vinyl or adhesive tape unless the ambient temperature is above 80°F (27°C). Attempting to peel the vinyl at lower temperatures will cause the material to tear and permanently damage the custom banner face. In cooler weather, warm the shop or gently heat the taped perimeter with a heat gun on low before peeling.

When taking down screens or preparing frames for graphic replacement:

- **Recommended timing (Defer to Band Camp):** It is usually much easier to leave the vinyl on the frames over the winter and wait until summer band camp of the following season before attempting removal. Cold late-fall temperatures following competition season make adhesive stiff and brittle, drastically increasing tear risk.
- **Solar heating advantage:** The warmer it is outside, the better. Staging the screens outdoors in direct sunlight for just a few minutes prior to peeling warms the conduit and softens the tape adhesive, allowing the vinyl and tape to release cleanly and effortlessly with minimal pull resistance.
- **Remove greenhouse snap clamps:** Carefully pry off all 16 greenhouse snap clamps from the perimeter tubing.
- **Peel slowly at shallow angle:** Starting at one corner, slowly peel the vinyl wrap back at a shallow angle. Warm adhesive releases cleanly from the bare EMT conduit without pulling the scrim or ink layer.
- **Tape stripping & cleanup:** Peel remaining tape strips from the conduit. Because the conduit was not aggressively degreased during initial assembly, tape strips will pull off cleanly with minimal adhesive residue.

## Appendix B Index

This index applies to Appendix B. Use the numbered references in the Appendix B component table to locate purchase sources, print material, and digital part files. In the digital manual, select a descriptive link. For a printed copy, the complete URL is shown beneath each link.

### Purchase sources

**[1] [3/4-in. x 10-ft EMT conduit - Home Depot](#)**

<https://www.homedepot.com/p/3-4-in-x-10-ft-Electrical-Metallic-Tubing-EMT-Conduit-0550110000/202068040>

**[2] [Corner brackets - Amazon](#)**

<https://www.amazon.com/dp/B0D5BBKCKG>

**[3] [T-brackets - Amazon](#)**

<https://www.amazon.com/dp/B0CKQW11SJ>

**[4] [#8 x 1/2-in. self-drilling screws - Home Depot](#)**

<https://www.homedepot.com/p/Teks-8-x-1-2-in-External-Hex-Flange-Hex-Head-Self-Drilling-Screws-280-Pack-21308/100188836>

**[5] [Black ASA filament - Amazon](#)**

<https://www.amazon.com/dp/B09TR84J5M>

**[8] [Greenhouse snap clamps \(1-in.\) - Amazon](#)**

<https://www.amazon.com/dp/B0BJJKZ5L?th=1>

**[9] [Heavy-duty double-sided tape - Amazon](#)**

<https://www.amazon.com/dp/B07BBL4JXJ/?th=1>

### Digital part files

**[6] Corner plug digital files**

**[FreeCAD source file](#)**

[https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/\\_Sideline%20Screen/Corner%20Plug.FCStd](https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/Corner%20Plug.FCStd)

**[STEP file](#)**

[https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/\\_Sideline%20Screen/STEP/Corner%20Plug-Part.step](https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/STEP/Corner%20Plug-Part.step)

**[3MF print file](#)**

[https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/\\_Sideline%20Screen/3MF/Corner%20Plug-Part.step.3mf](https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/3MF/Corner%20Plug-Part.step.3mf)

**[7] Hinged arm clip digital files**

**[FreeCAD source file](#)**

[https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/\\_Sideline%20Screen/Hinged%20Arm%20Clip.FCStd](https://github.com/ericrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/Hinged%20Arm%20Clip.FCStd)

[STEP file](#)

[https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/\\_Sideline%20Screen/STEP/Hinged%20Arm%20Clip-Part001.step](https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/STEP/Hinged%20Arm%20Clip-Part001.step)

[3MF print file](#)

[https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/\\_Sideline%20Screen/3MF/Hinged%20Arm%20Clip-Part001.3mf](https://github.com/ericcrowe/music/blob/7ab68c66b4ecbd72d9e039212acab02a002df1bb/PCHSMB/_Sideline%20Screen/3MF/Hinged%20Arm%20Clip-Part001.3mf)

**[11]** [2026 Colorado Bandmasters Association \(CBA\) Marching Band Rulebook \(PDF\)](#)

[https://ebf5c7e8-3869-4325-bd98-ce10f57d7545.filesusr.com/ugd/83f67e\\_cb5fe6efdbdb4ce98f619580f694bc94.pdf](https://ebf5c7e8-3869-4325-bd98-ce10f57d7545.filesusr.com/ugd/83f67e_cb5fe6efdbdb4ce98f619580f694bc94.pdf)